KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI
COLLEGE OF SCIENCE
FACULTY OF PHYSICAL AND COMPUTATIONAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE

Course Outline, Teaching and Learning Schedule

CSM 399 Web-Based Concepts and Development

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Lecture Periods: 10:30-15:00 Monday
Office Hours: 12:30-14:30 Tuesdays
Contact Number: 0543286900

Course Description:
This course focuses on designing and developing Web-based applications using a variety of programming languages and tools. We will learn to create web pages using XHTML and CSS. Topics will include the basics of HTML, tables and forms. We will learn design concepts in the cascading style sheet and venture into scripting with Java scripts. We will briefly look at React JS and have a capstone project presentation at the end of the course.

Course Learning Outcomes:

By the end of the course the student should be able to:

- Understand the challenges, technologies and issues in developing and deploying Web-based applications
- Build and deploy web applications in line with current trends in technology
- Apply a systems approach to business problems
- Be part of a team that helps solve business problems using technology
- Improve their oral, written and group communication skills with in-class participation and final group presentation

Course Content Outline

Unit	Topics	Sub-topics (if any)	Week(s)
1	Introduction HTML	• Introduction and course overview • Blended learning and Netiquette • HTML • Document Object Model • Common HTML tags • Designing simple web pages	2
2	CSS	• Types of styling • Responsive designing	2

		• Using Bootstrap • SASS	
3	JavaScript	• Events • Variables • DOM Manipulation • Intervals • APIs	2
4	ReactJS	• Creating a React App • Component Structure • Handling Events • Lifecycle Methods and Hooks • Forms and Inputs • Lists and Keys • Context API • Router • Error Boundaries	3
5	Capstone Project	Group presentations	1

Mode(s) of Delivery:
This course will be using a blended approach. Approximately 40% of the lectures will be conducted via Microsoft Teams, and the remainder will be held face-to-face. All course material will be made available on the VClass.

Assessment Policy:
Each student is expected to attend at least 80% of the lectures, and this will be checked by the attendance taken in each class. There will be a series of quizzes on the VClass at agreed-upon times.
The various components will be graded as follows:

- **Continuous Assessment - 30%**
  - Attendance – 5%
  - Quizzes – 10%
  - Midsemester Exam – 15%
- **End of semester Exam - 70%**
  - Capstone Project – 30%
  - Final Exam – 40%

Required Textbooks:
We will officially not use a textbook for this course. However, materials such as lecture notes, PowerPoint slides, and other materials from the Internet will be provided.

Other Reading Textbooks/Resources:

- Duckett, John (2014), Web Design with HTML, CSS, JavaScript and jQuery Set, Wiley
- Purewal, Samuel (2014), Learning Web App Development, O'Reilly
- Dennis, Allan; Wixom, Barbara; Roth, Roberta M., (2018), Systems Analysis and Design, 7th Edition, Wiley

Capstone Project Instructions

- **Objective:**

Design, develop, and deploy a fully functional web application that integrates concepts from HTML, CSS, JavaScript, and ReactJS. This capstone project should simulate a real-world industry solution, requiring collaboration, role distribution, and adherence to software development best practices.

- **Project Overview:**

Each team should create a web-based solution addressing a real-world problem. The project must demonstrate proficiency in responsive design, dynamic functionality, state management, and effective UI/UX principles.

- **Roles and Responsibilities:**

Each team will assign members to the following roles:

1. Project Manager (PM)
   - Coordinates the team, sets deadlines, and ensures tasks are completed on time.
   - Organizes meetings and communicates progress with the instructor.
   - Oversees documentation and ensures deliverables are submitted.

2. Front-End Developer
   - Designs and implements the user interface using HTML, CSS (and optionally ReactJS).
   - Creates a responsive, accessible, and visually appealing layout.
   - Implements dynamic components and state management.

3. Back-End Developer (if applicable)
   - Develops or simulates APIs and server-side logic (if a mock server is used).
   - Manages data storage, retrieval, and processing. (Node.js or a lightweight JSON file can be used.)

4. UI/UX Designer
   - Focuses on user experience and interface consistency.
   - Creates wireframes or mock-ups and ensures intuitive navigation.

5. Quality Assurance (QA) Engineer
   - Tests the application for bugs, performance, and cross-browser compatibility.
   - Ensures the project meets accessibility and responsiveness standards.

6. Each group is allowed to define other roles they feel are applicable and define specific responsibilities for those roles.

- **Deliverables and Milestones:**

1. Project Proposal (Due by Friday 6th February 2026)
- A one-page document outlining:
  - Problem statement
  - Target audience
  - Core features
  - Assigned roles

2. Design Document (Due by Friday 6th March 2026)
- Wireframes or mock-ups for the project.
- Colour scheme, typography, and overall design choices.

3. Development Phase (Weekly updates will be required from PMs)
- Implementation of project features in stages:
  ****The group should come up with their own project management milestones

4. Testing and Review (Depends on the group)
- Each team will conduct peer reviews and bug testing.
- Documentation of issues found and resolved.

5. Final Presentation (Monday 30th March 2026. Specific Timeslots will be communicated shortly)
- A live demonstration of the project.
- Explanation of roles, technologies used, and challenges faced.

- **Technical Requirements**

- Frontend: HTML, CSS, JavaScript (and optionally, ReactJS).
- Optional Backend: Mock APIs using JSON server or Node.js.
- Responsive Design: The application must be fully functional on mobile, tablet, and desktop screens.
- Version Control: Use Git for collaboration and maintain a structured repository.

- **Capstone Project Assessment Criteria**

Criteria	Description	Weight
Functionality	The application works as intended without errors or crashes	30
Design and UI/UX	The application is visually appealing, responsive, and easy to use	20
Code Quality	Clean, modular, and well-commented code following best practices	20
Team Collaboration	Effective teamwork and role fulfilment	15
Presentation	Clear and engaging presentation demonstrating key features	15

Academic Integrity

The Academic Code of Conduct and Ethics prohibits certain acts of misconduct by students enrolled at KNUST. Other University policies and procedures may also apply. Depending on the circumstances, the acts of misconduct described below may be considered a violation or an infraction.

Violations

The following acts are examples of violations.

- Behaviour during Assessments: any intentional giving or use of external assistance during an assessment
- Fabrication: any falsification or invention of data, citation or other authority in an academic exercise.
- Plagiarism: any passing off of another's ideas, words, or work as one's own.
- Unauthorised Collaboration: collaboration in any academic exercise unless your instructor has stated that such collaboration is permitted.
- Theft of Resource Materials: any unauthorised copying of resources such as course materials, library materials, computer software, or media productions.
- Previously Submitted Work: presenting work prepared for and submitted to another course.
- Unauthorised Access: any unauthorised access to any person's files or computer account.
- Unauthorised Research: failure to obtain approval of the Institutional Review Board for research involving human subjects.
- Alteration or Misuse of University documents: any alteration or misuse of university documents, including acts of forgery and/or furnishing false information.
- Disruption of Academic Activity: disruptive behaviour, wilful disobedience, profanity or vulgarity which continues after a warning has been given.
- Acts or Threats of Damage to Property: acts or threats of abuse, misuse, damage, or destruction of property belonging to or located on university-controlled property or facilities.
- Violation Defined by your Instructor: any other intentional violation of rules or policies established by a course faculty member/academic supervisor.